

ECE595: Power Converters for AC Systems

Lecture 26 (1-ph and 3-ph B2B Configs.)

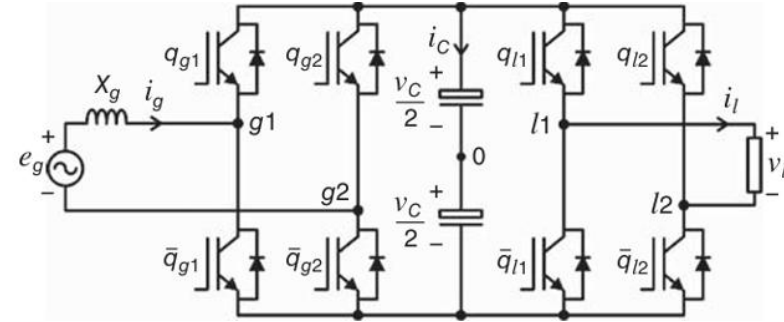
Prof. Euzeli Dos Santos

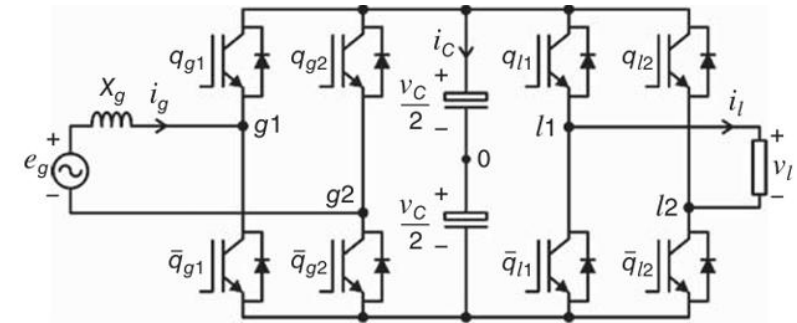


Problem

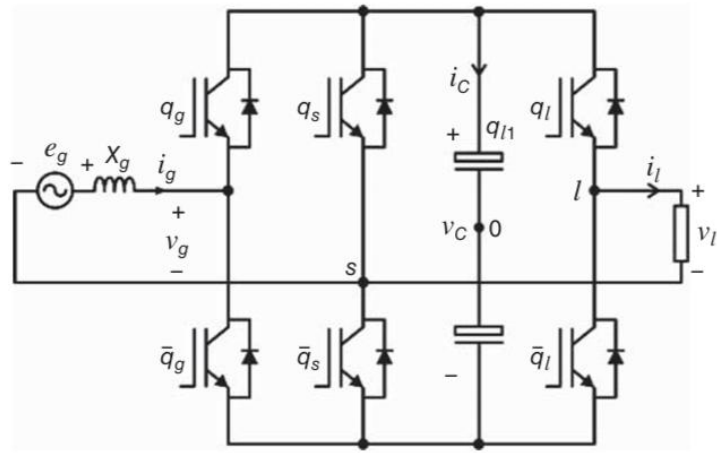
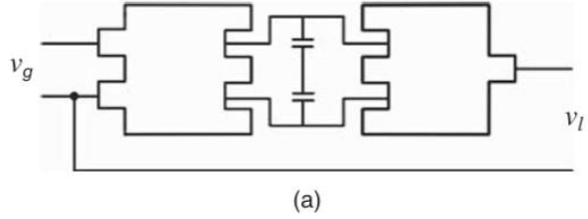
For the single-phase B2B configuration aside determine the minimum DC-link voltage if the grid voltage is 120Vrms (60 Hz) and the grid inductance is $\frac{10}{2\pi} \text{ mH}$ (disregard resistance).

The expected voltage at the load is 120Vrms and assume a grid current of 20 Arms.

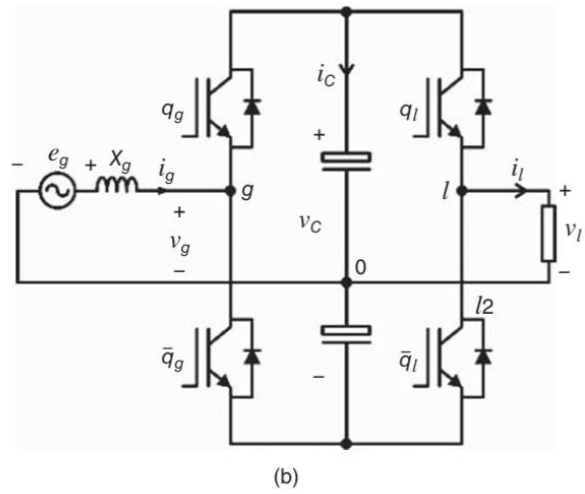
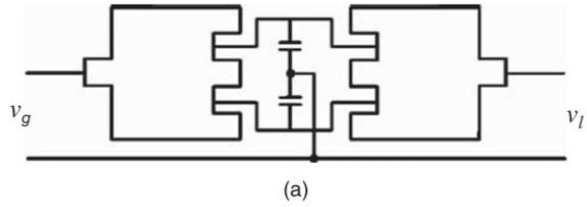




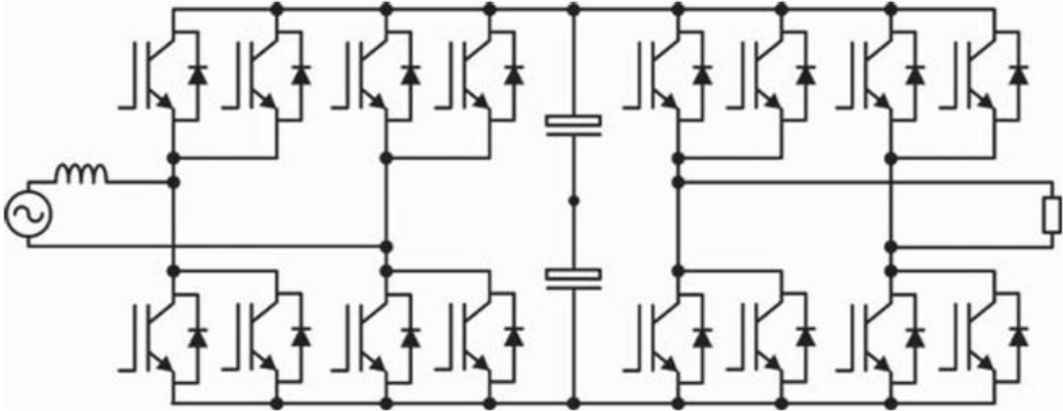
Topology With Component Count Reduction (Shared-leg)



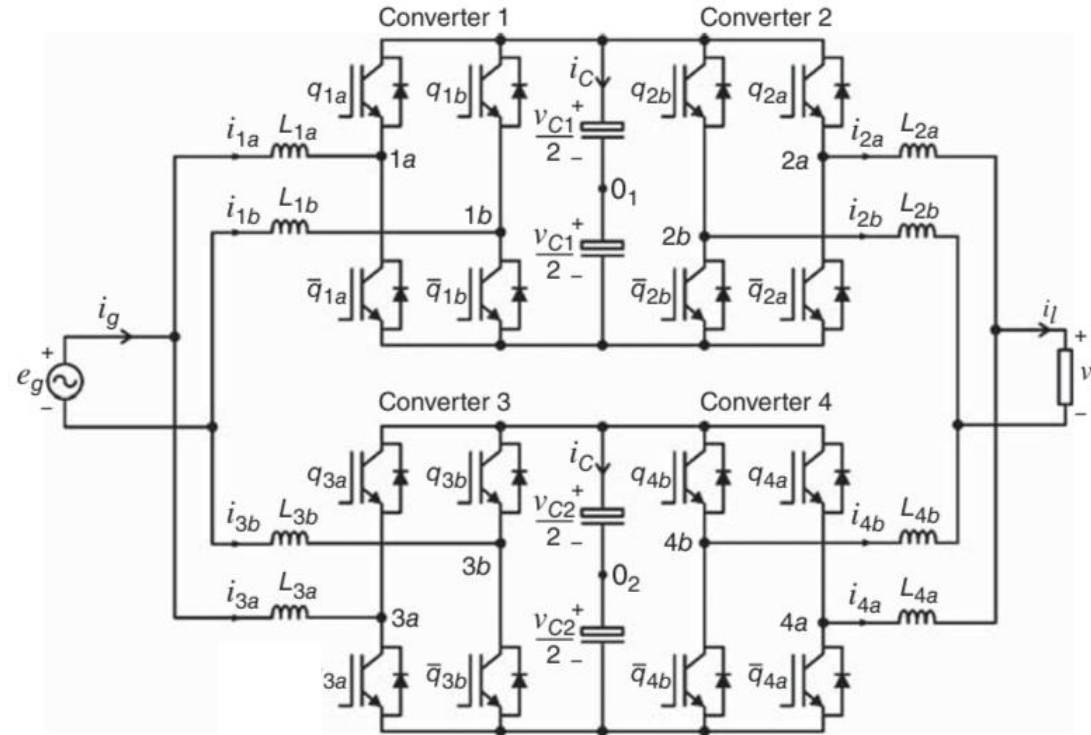
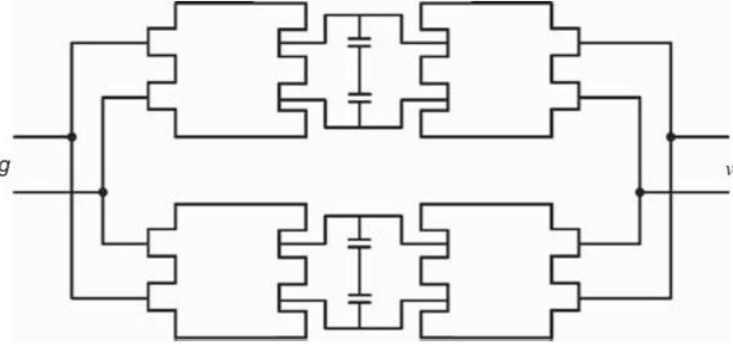
Topology With Component Count Reduction (Half-bridge)

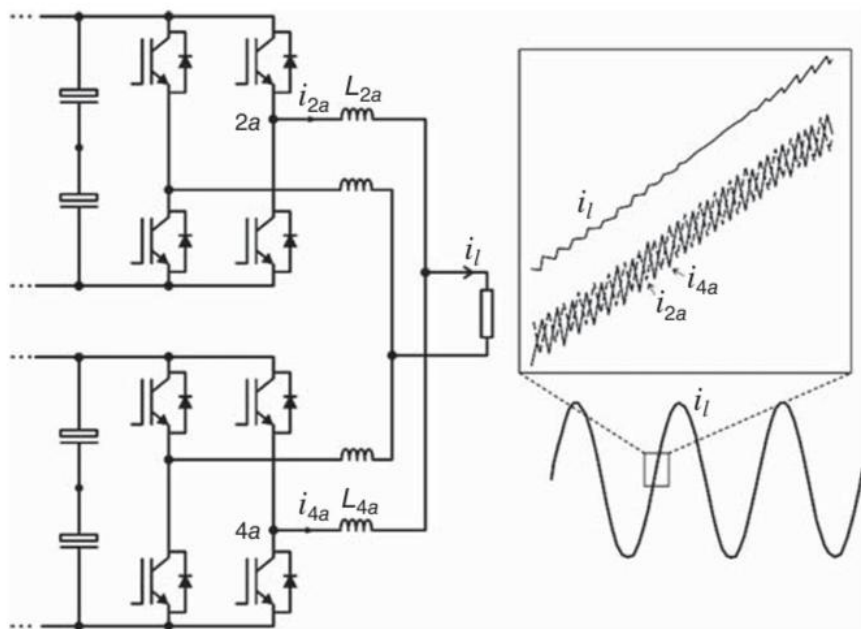


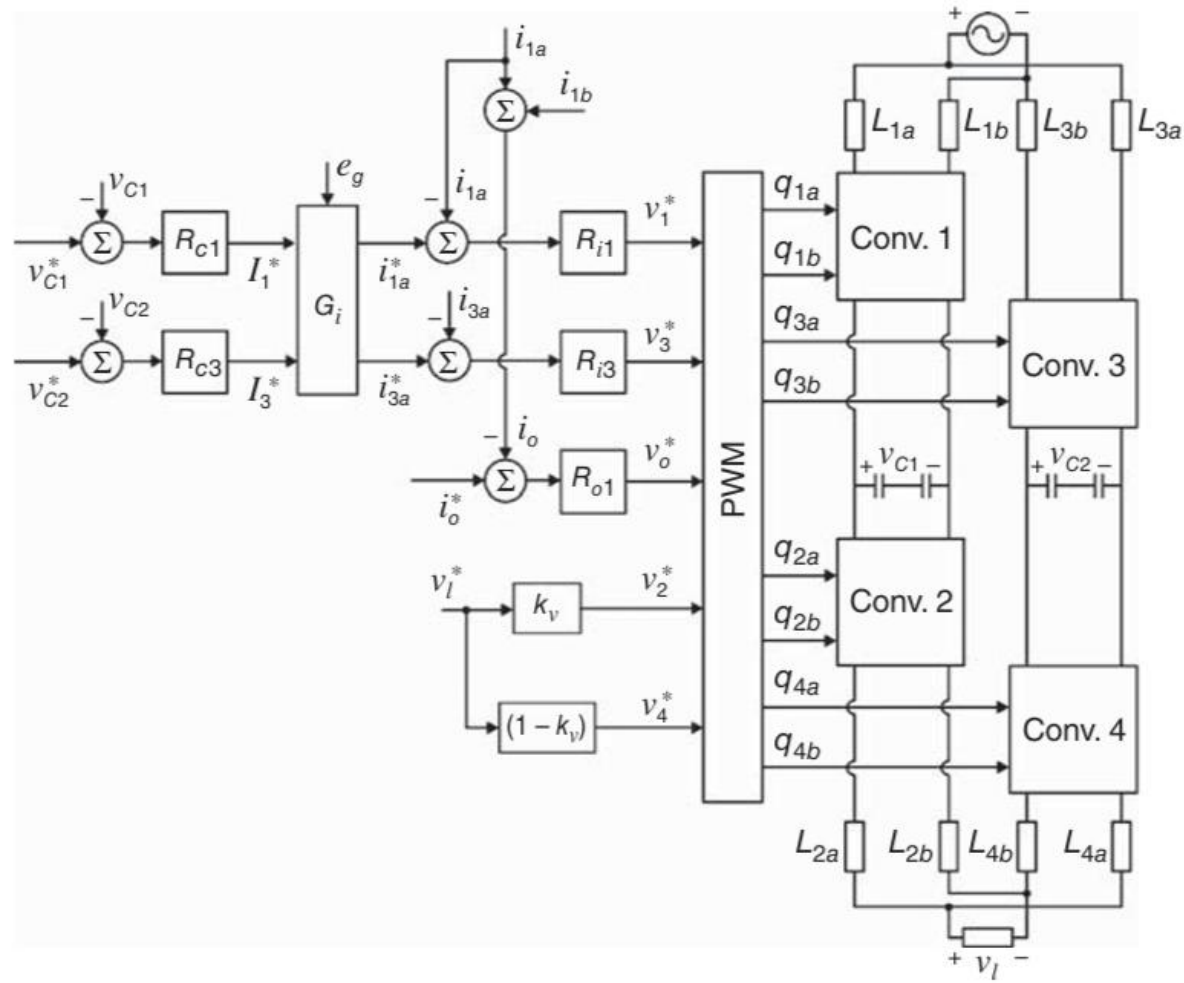
Topology With Increased Number of Switches



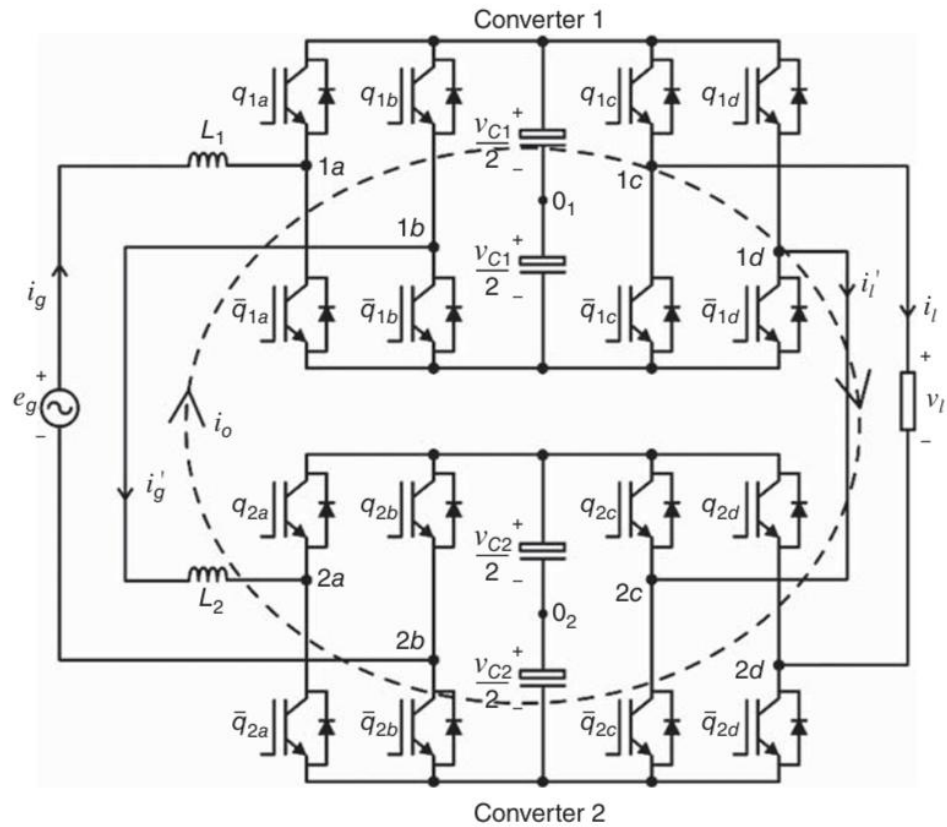
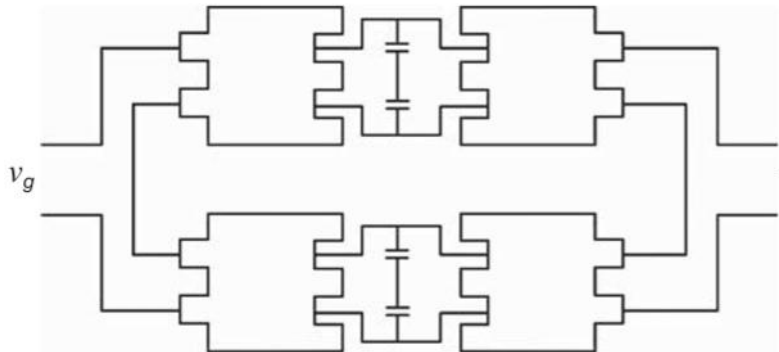
Topology With Increased Number of Switches (Converters in Parallel)

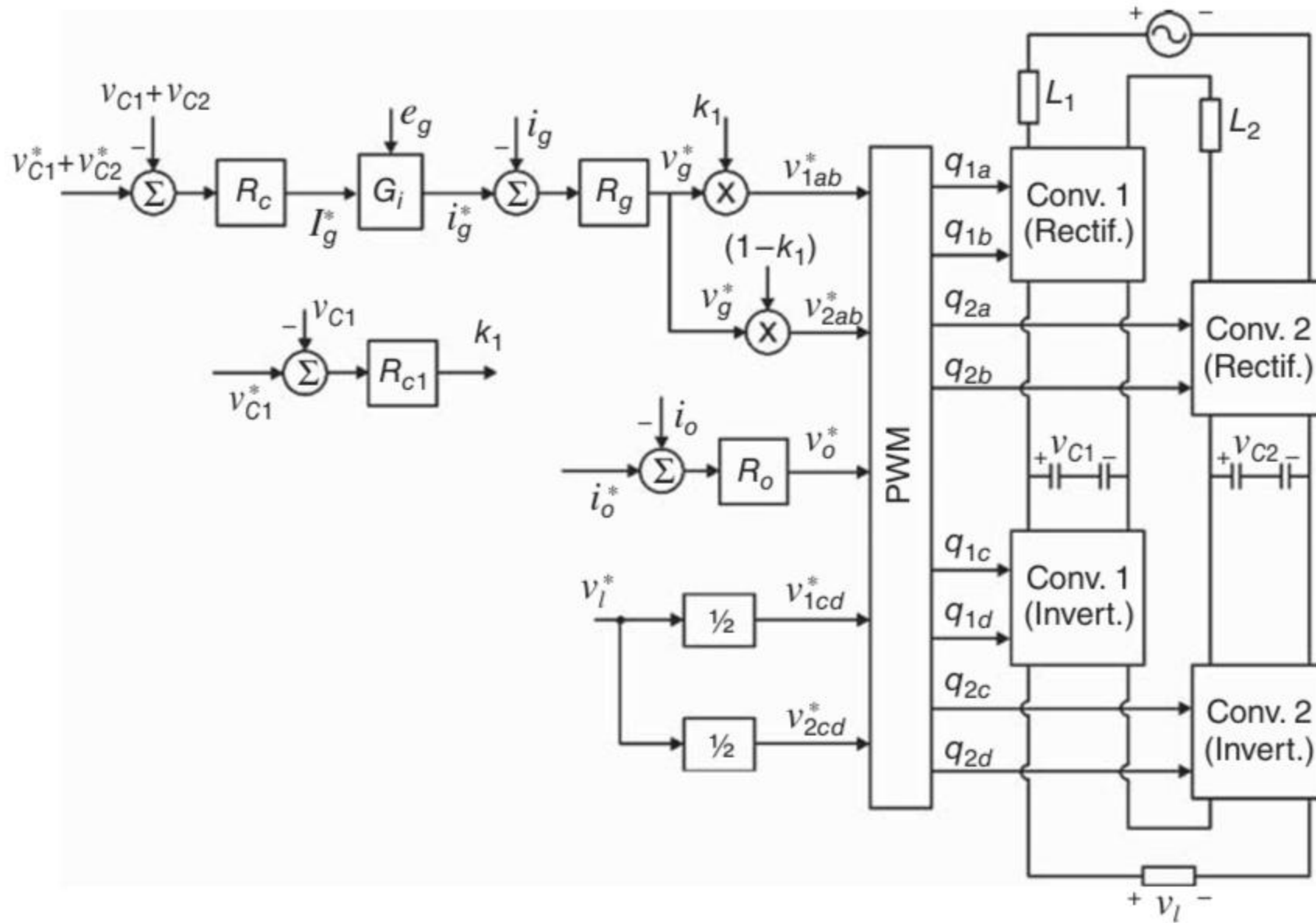




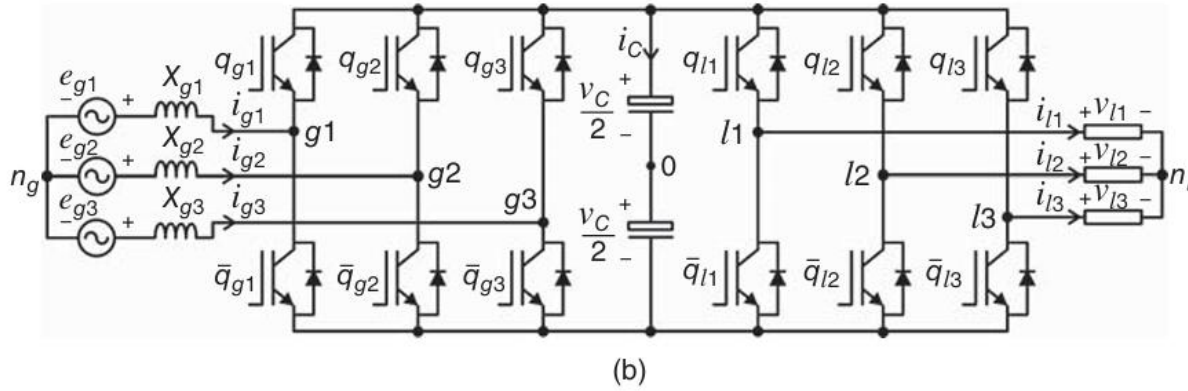
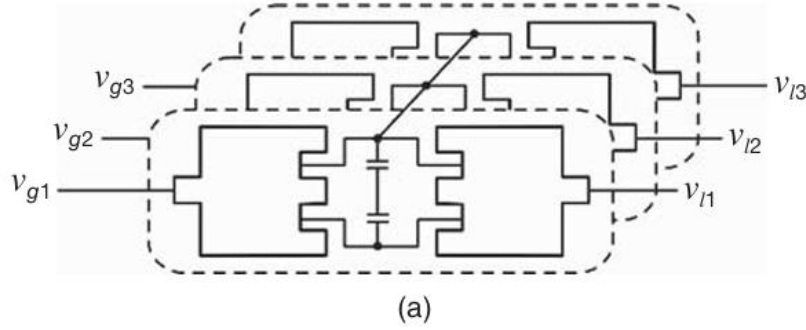


Topology With Increased Number of Switches (Converters in Series)





Three-phase to Three-phase (Full-Bridge)



$$v_{g1} = e_{g1} - r_g i_{g1} - l_g \frac{di_{g1}}{dt}$$

$$v_{g2} = e_{g2} - r_g i_{g2} - l_g \frac{di_{g2}}{dt}$$

$$v_{g3} = e_{g3} - r_g i_{g3} - l_g \frac{di_{g3}}{dt}$$

$$v_c = \frac{1}{C} \int (q_{g1} i_{g1} + q_{g2} i_{g2} + q_{g3} i_{g3} - q_{l1} i_{l1} - q_{l2} i_{l2} - q_{l3} i_{l3}) dt$$

$$v_{g1} = v_{g10} - v_{n_g0}$$

$$v_{g2} = v_{g20} - v_{n_g0}$$

$$v_{g3} = v_{g30} - v_{n_g0}$$

$$v_{l1} = v_{l10} - v_{n_l0}$$

$$v_{l2} = v_{l20} - v_{n_l0}$$

$$v_{l3} = v_{l30} - v_{n_l0}$$

$$v_{g10} = (2q_{g1} - 1) \frac{v_C}{2}$$

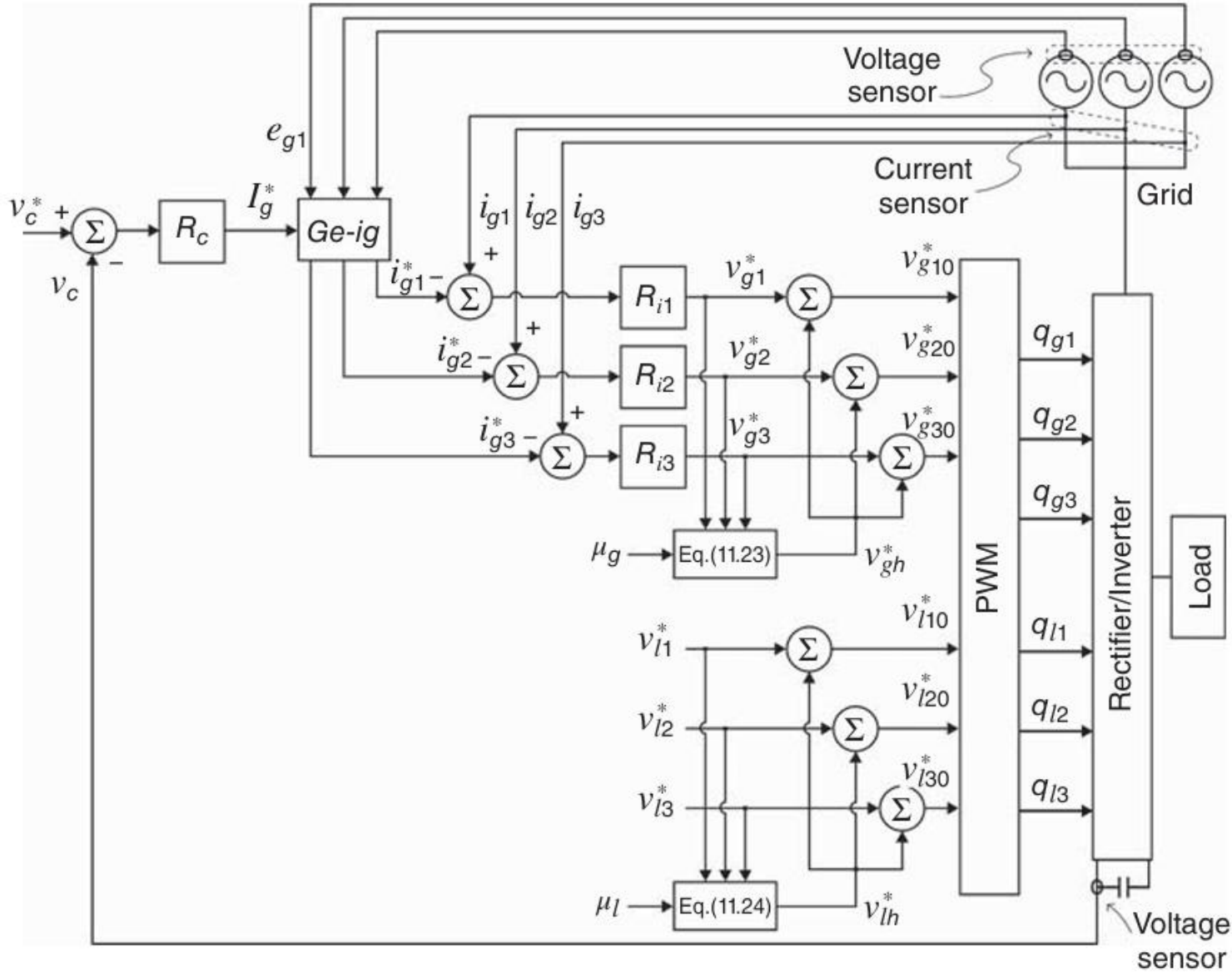
$$v_{g20} = (2q_{g2} - 1) \frac{v_C}{2}$$

$$v_{g30} = (2q_{g3} - 1) \frac{v_C}{2}$$

$$v_{l10} = (2q_{l1} - 1) \frac{v_C}{2}$$

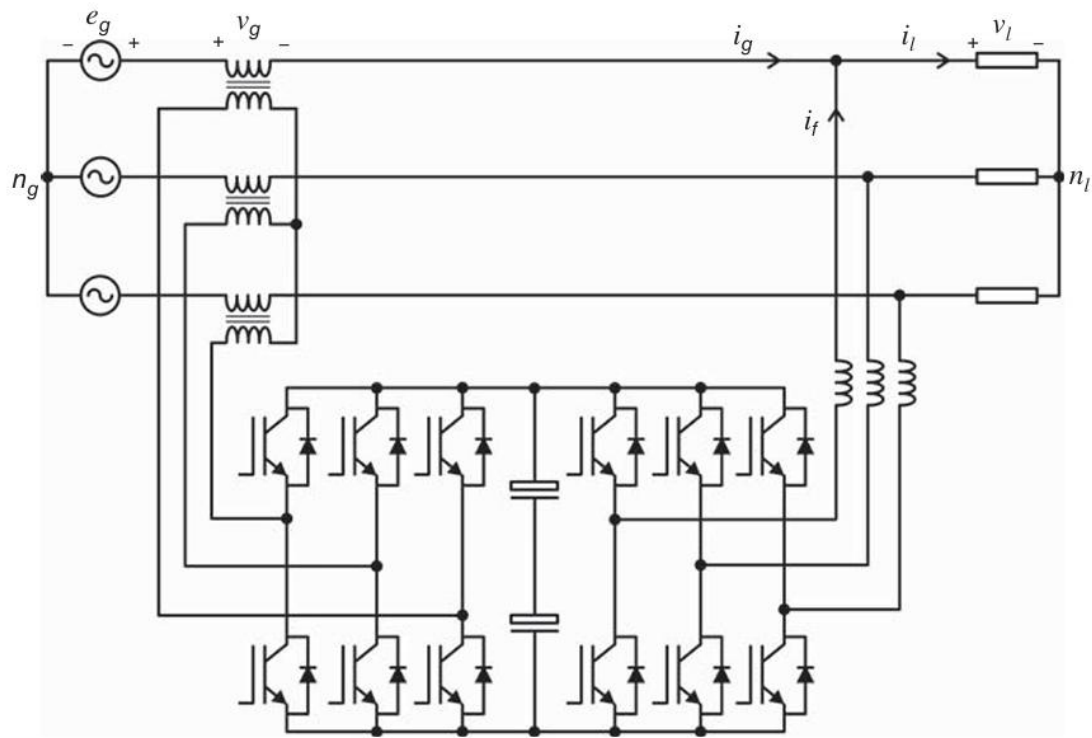
$$v_{l20} = (2q_{l2} - 1) \frac{v_C}{2}$$

$$v_{l30} = (2q_{l3} - 1) \frac{v_C}{2}$$

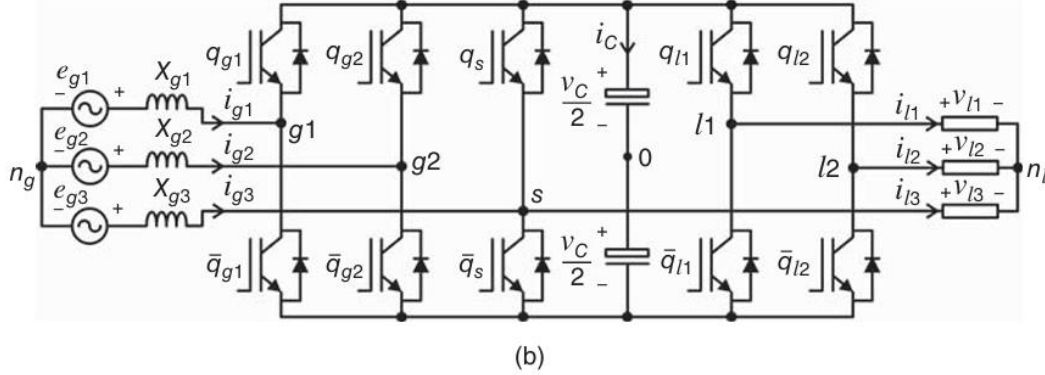
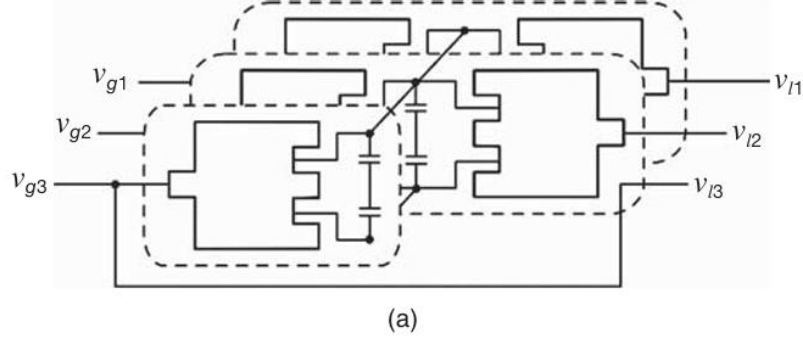


$$\begin{aligned}
 v_{g10}^* &= v_{g1}^* + v_{gh}^* \\
 v_{g20}^* &= v_{g2}^* + v_{gh}^* \\
 v_{g30}^* &= v_{g3}^* + v_{gh}^* \\
 v_{l10}^* &= v_{l1}^* + v_{lh}^* \\
 v_{l20}^* &= v_{l2}^* + v_{lh}^* \\
 v_{l30}^* &= v_{l3}^* + v_{lh}^*
 \end{aligned}$$

Application (Universal Active Power Filter)



Topology with component count Reduction (shared leg)



$$v_c = \frac{1}{C} \int [q_{g1}i_{g1} + q_{g2}i_{g2} + q_s(i_{g3} - i_{l3}) - q_{l1}i_{l1} - q_{l2}i_{l2}]dt$$

$$v_{g10}^* = v_{g1}^* - v_{g3}^* + v_h^*$$

$$v_{g20}^* = v_{g2}^* - v_{g3}^* + v_h^*$$

$$v_{s0}^* = v_h^*$$

$$v_{l10}^* = v_{l1}^* - v_{l3}^* + v_h^*$$

$$v_{l20}^* = v_{l2}^* - v_{l3}^* + v_h^*$$

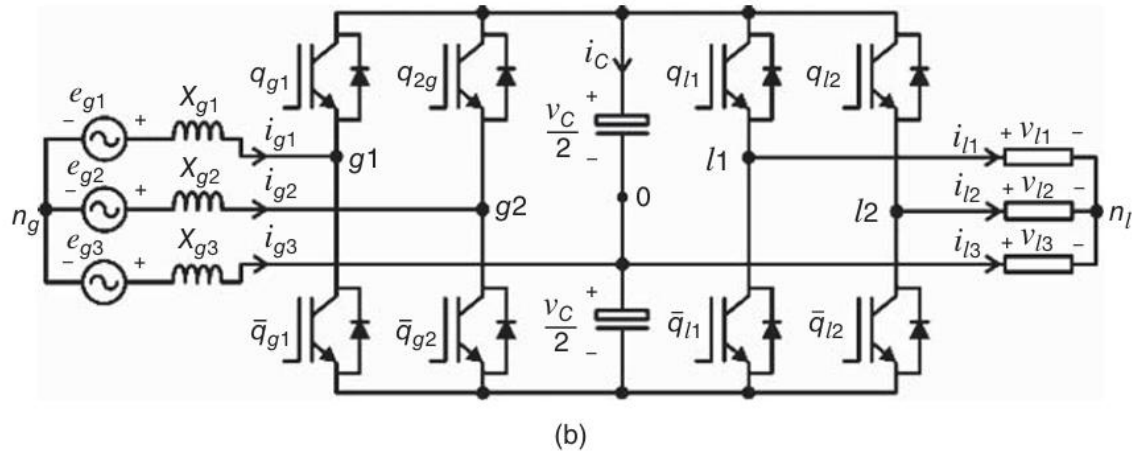
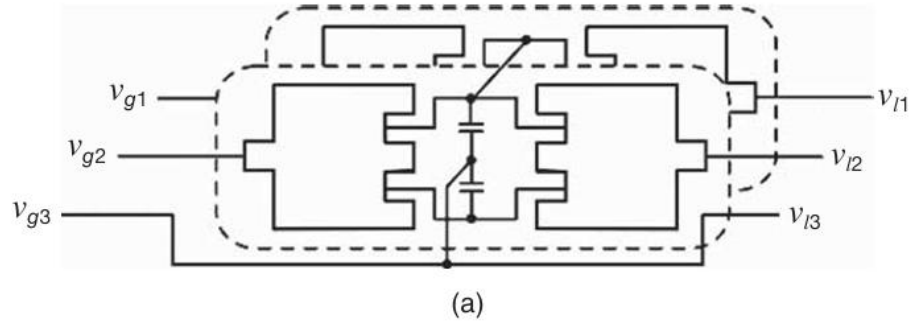
11.3.3 dc-link Voltage Requirement

To guarantee the same input and output voltages as in the full-bridge converter, the minimum dc-link voltage for the shared leg topology must obey the following requirement:

$$v_c \geq \max\{v_{gx0}^* - v_{gy0}^*, v_{gx0}^* - v_{s0}^*, v_{lx0}^* - v_{ly0}^*, v_{lx0}^* - v_{s0}^*, v_{gx0}^* - v_{ly0}^*\} \quad (44)$$

with $x = 1, 2$; $y = 1, 2$; and $x \neq y$.

Topology with component count Reduction (Half-bridge)



Topology with switches in parallel

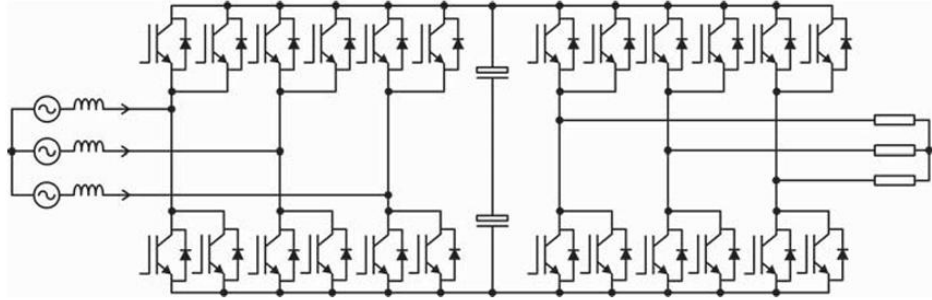
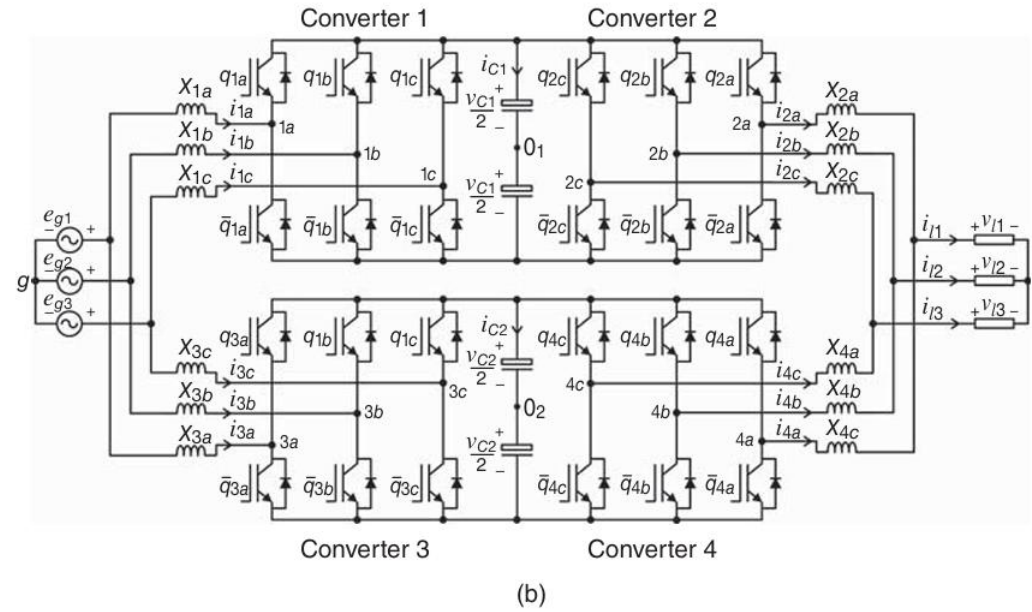
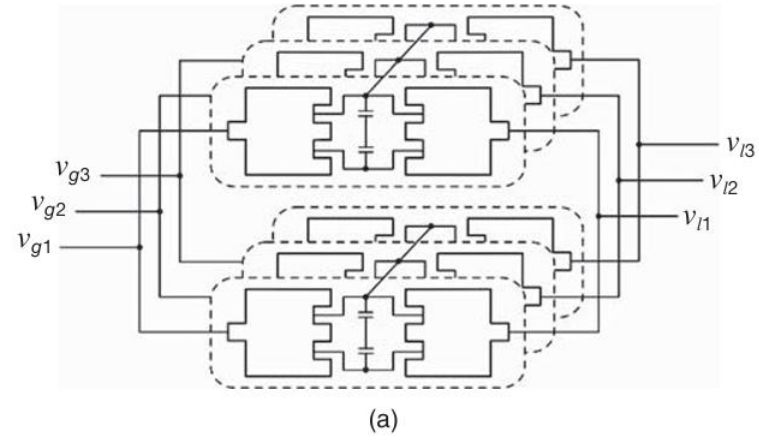


Figure 11.5 Three-phase back-to-back converter with switches in parallel.



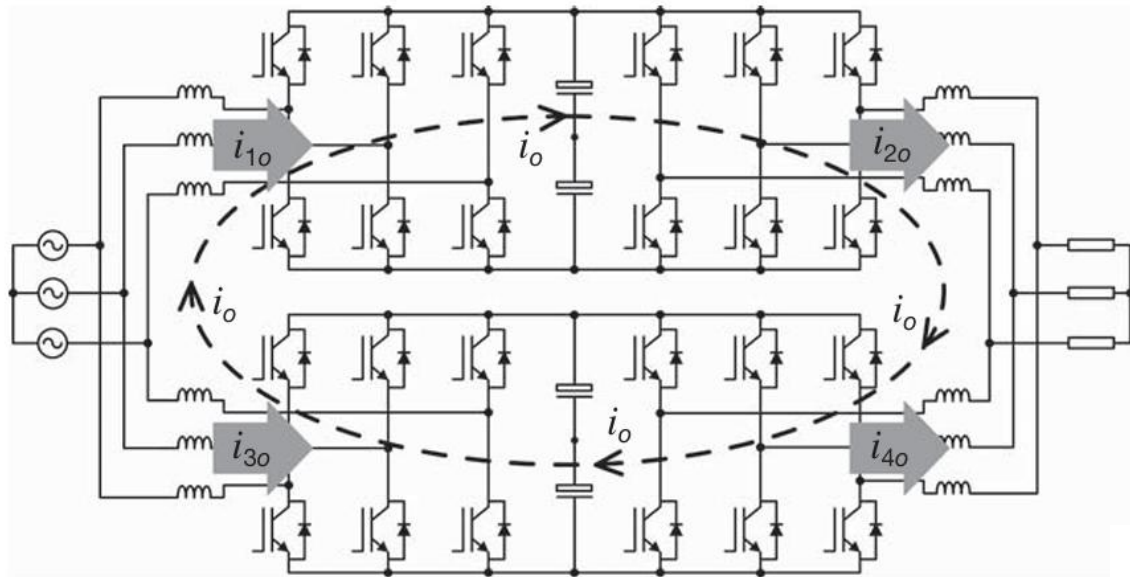
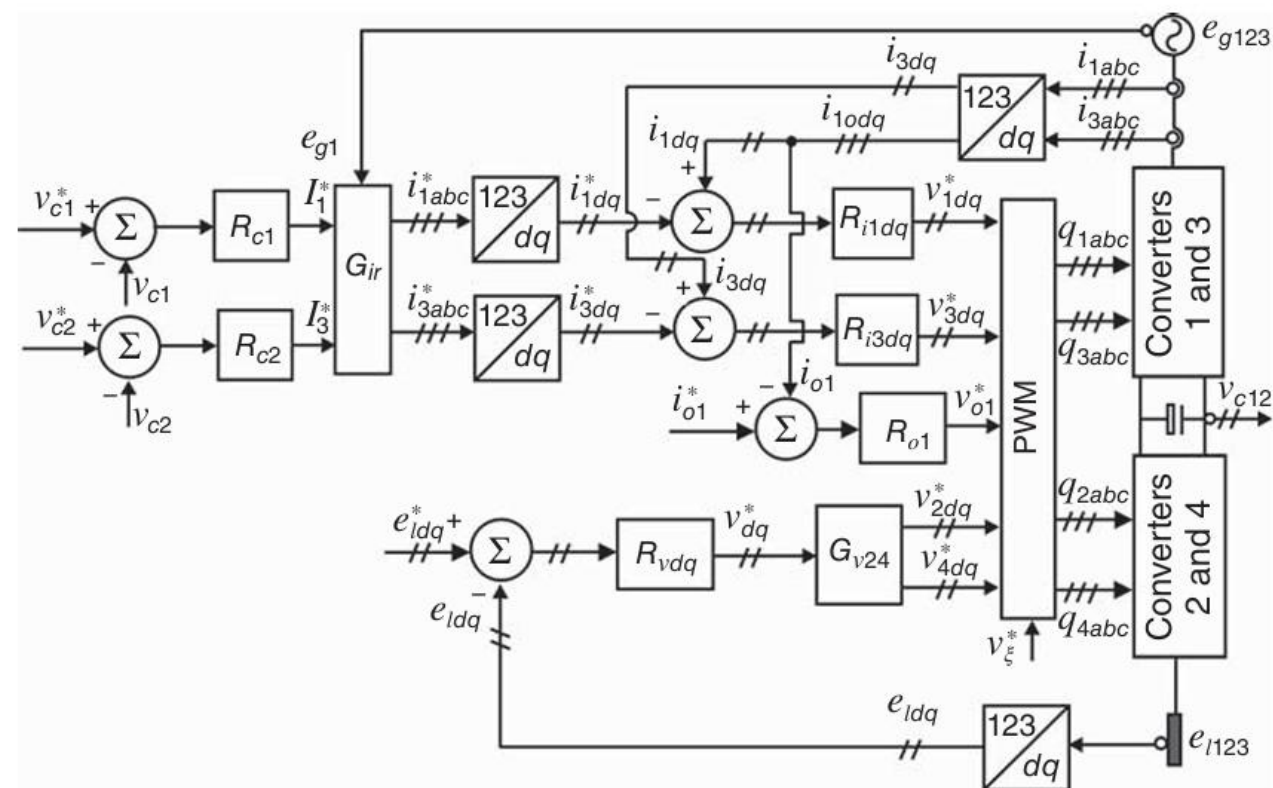
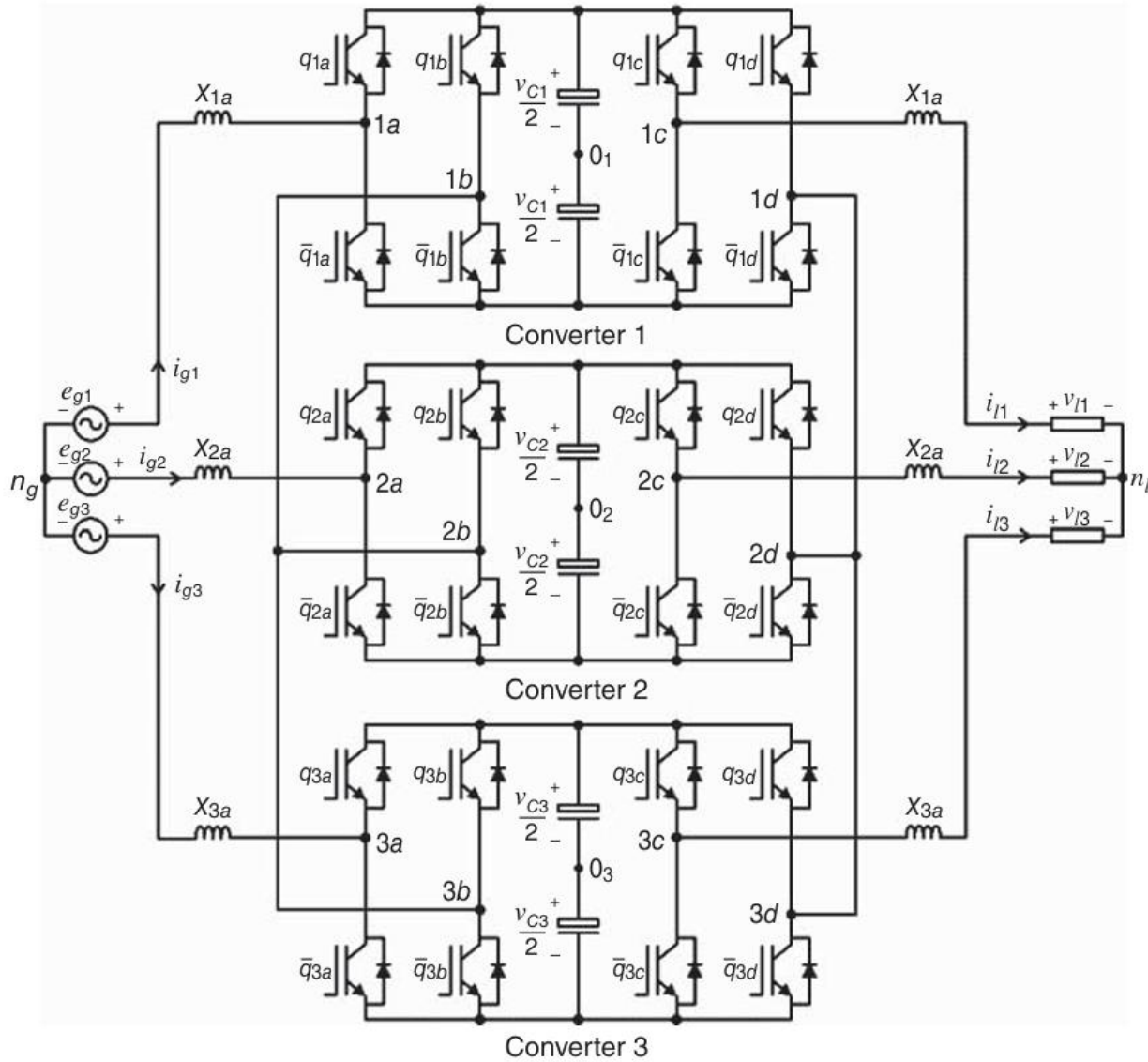


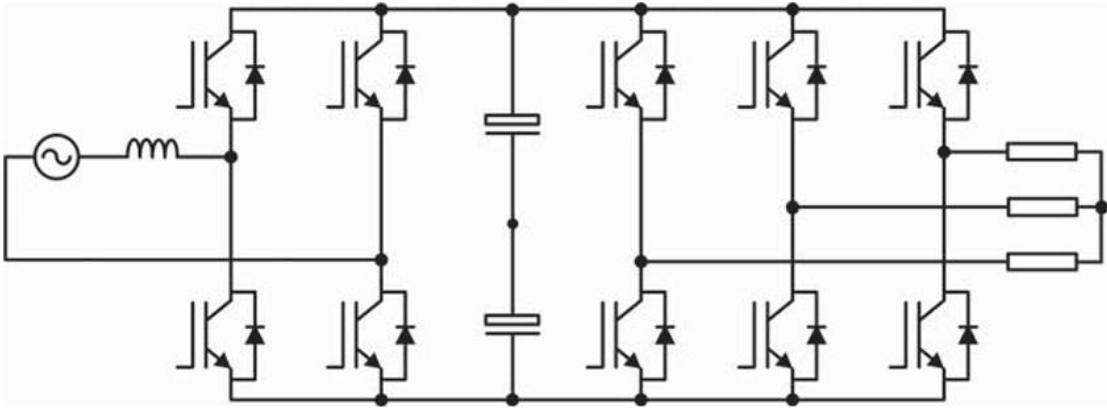
Figure 11.7 Circulating current between both back-to-back converters in parallel.



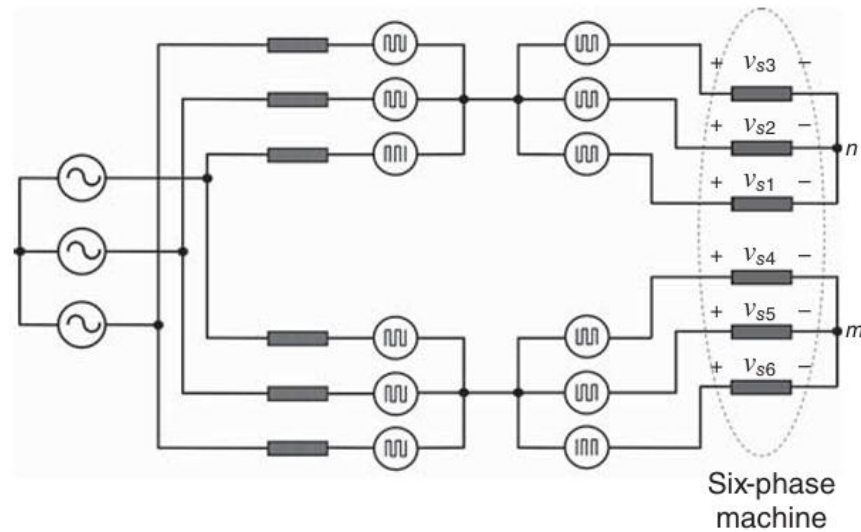
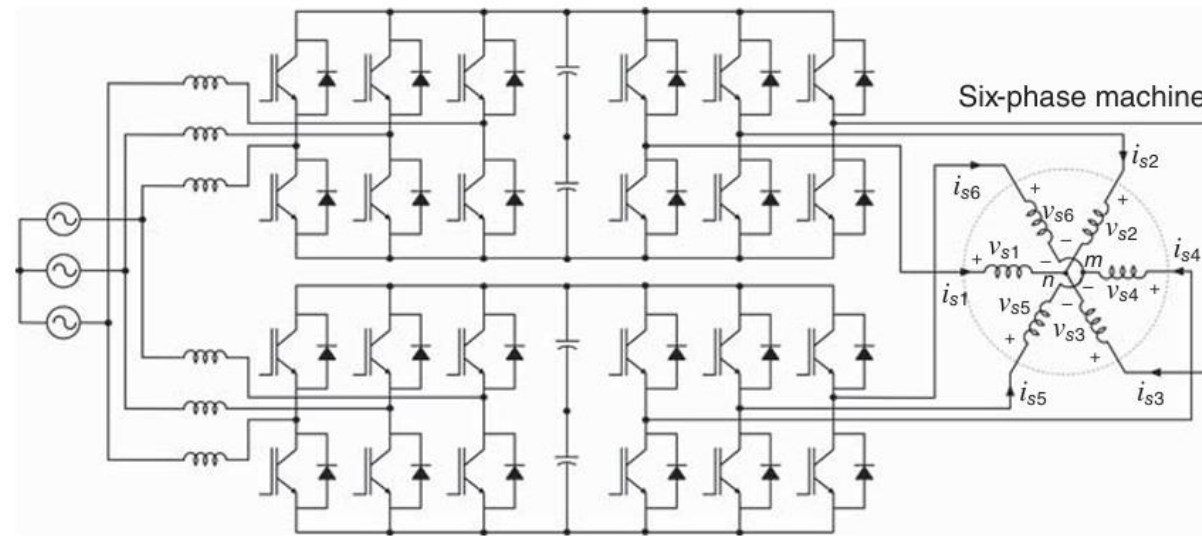
Other connections



Other connections

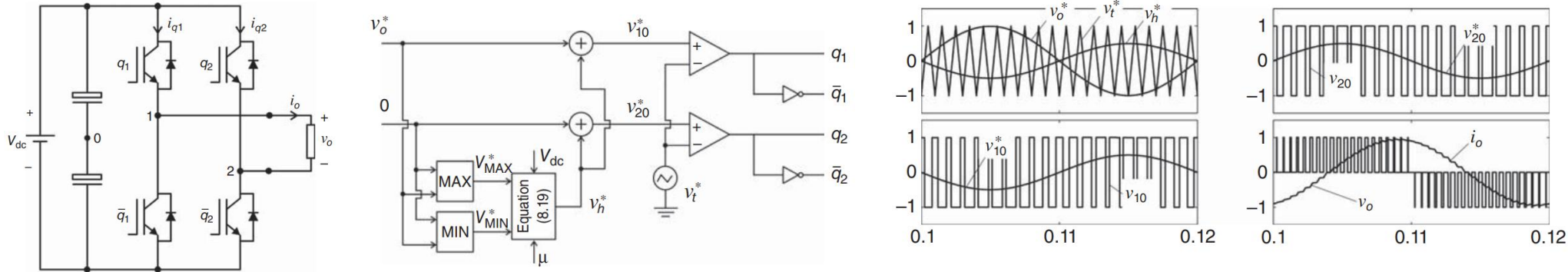


Other connections



Problem 1

The single-phase converter and its analog PWM implementation is shown below, all the waveforms are presented for the case with $\mu=0.5$.



- Determine the load peak current if $v_{10}^* = \sin(\omega t)$ and $v_{20}^* = 0$, where ω is the same frequency of the v_o^* .
- Repeat (a) if $v_{10}^* = 0$ and $v_{20}^* = -\sin(\omega t)$.
- Repeat (a) if $v_{10}^* = -0.25\sin(\omega t)$ and $v_{20}^* = 0.75\sin(\omega t)$.

Problem 2

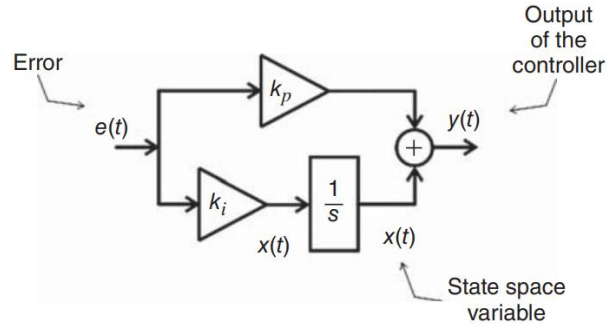
For the MATLAB code below, make changes such that you feed the RL load with a half-bridge single-phase converter controlled by hysteresis.

```
%Simulation of a RL load connected to a Controlled-
Voltage-Source
R=5;      %Resistance of the RL load
L=0.05;   %Inductance of the RL load
io=0;     %state variable (load current)
vo=100;   %input variable (load voltage)
h=1e-4;   %step size
j=0;      %variable used to storage the outcomes
t=0;      %initial value of the digital time
tmax=1;   %maximum simulation time

while t<=tmax, %begining of the simulation loop
    t = t + h; %time incrementation
    io = io + (vo-R*io)*h/L; %state space variable 1
    % storage of the variables
    j = j + 1;
    time(j)=t;
    current(j)=io;
    voltage(j)=vo;
end
figure(1),plot(time,voltage)
figure(2),plot(time,current)
```


Problem 3

Repeat Problem 2 by changing the hysteresis control to PI controller, see below.



```
%Simulation of a RL load connected to a Controlled-
Voltage-Source
R=5;      %Resistance of the RL load
L=0.05;   %Inductance of the RL load
io=0;     %state variable (load current)
vo=100;   %input variable (load voltage)
h=1e-4;   %step size
j=0;      %variable used to storage the outcomes
t=0;      %initial value of the digital time
tmax=1;   %maximum simulation time

while t<=tmax, %begining of the simulation loop
    t = t + h; %time incrementation
    io = io + (vo-R*io)*h/L; %state space variable 1
    % storage of the variables
    j = j + 1;
    time(j)=t;
    current(j)=io;
    voltage(j)=vo;
end
figure(1),plot(time,voltage)
figure(2),plot(time,current)
```

Problem 4

Below is presented a single-phase B2B topology and its control strategy. Propose changes in the block diagram aside if the input is a half-bridge rectifier, everything else remains the same.

